HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained robust Java Spring Boot microservices, building RESTful APIs that handled over 5,000 requests per minute for internal data processing workflows.
- Implemented secure data access patterns for enterprise AI capabilities, ensuring adherence to least-privilege principles and integrating with the company's Identity and Access Management (IAM) systems.
- Engineered Python scripts to automate data ingestion and validation for AI model training pipelines, processing approximately 1.5 TB of raw data weekly.
- Designed and implemented AI guardrails and prompt-injection mitigation strategies for GenAI applications, reducing the occurrence of unauthorized outputs by an estimated 40%.
- Integrated AI applications with SIEM platforms, ensuring comprehensive logging of security events and facilitating real-time threat detection and analysis.
- Collaborated with cross-functional teams including Cybersecurity, Data Engineering, and Legal/Privacy to embed security controls throughout the AI development lifecycle.
- Containerized AI services using **Docker** and deployed them to a managed Kubernetes cluster, configuring resource limits and health checks for reliable runtime operation.

SKILLS

AI Security – AI guardrails, prompt-injection mitigation, jailbreak protection, secure tool calling, input/output validation, sensitive data protection, adversarial testing, threat modeling

Application & API Security – API security, WAF, gateway security, SAST, DAST, SCA, vulnerability management, secrets management, secure SDLC

Cloud & Infrastructure – AWS, EC2, S3, Lambda, CloudWatch, Docker, Kubernetes, CI/CD, Jenkins, Git

Programming Languages – Java, Spring Boot, Python, SQL, Bash

Data & Analytics – Data pipelines, ETL, Splunk, Dynatrace, data modeling, Power BI, R, pandas, NumPy, scikit-learn

Identity & Access Management – IAM, least-privilege access, access controls, authentication, authorization

PROJECTS

AI Application Security Controls Implementation

- Developed and deployed input validation filters for an LLM-based customer service chatbot, mitigating approximately 85% of identified prompt injection attempts.
- Integrated a content safety API to scan user prompts and AI responses, flagging and blocking potentially harmful or inappropriate content before delivery.
- Authored technical documentation outlining secure coding practices for AI applications, which was adopted by the engineering team for new feature development.

Secure Data Pipeline for AI Feature Engineering

- Built a Python data pipeline on AWS Lambda to extract, transform, and load sensitive customer data for feature engineering, ensuring all data was encrypted at rest and in transit.
- Implemented role-based access controls within the data pipeline, granting access only to authorized personnel and AI services based on least-privilege principles.
- Designed and implemented automated data quality checks within the pipeline, identifying and flagging over 500 malformed data records per week before they reached model training.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)